



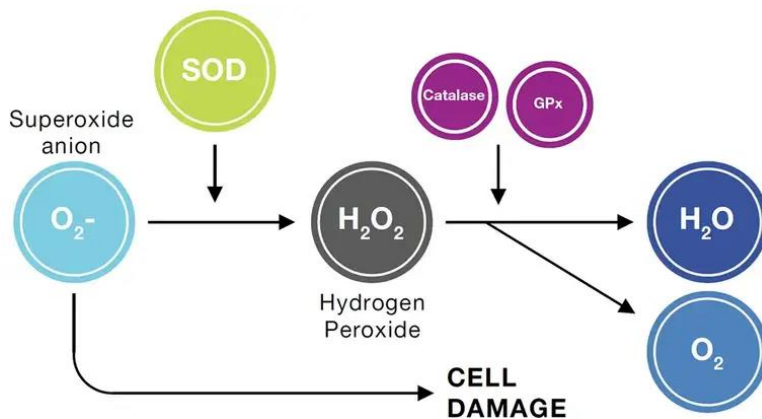
ANTIOXIDANT TESTING

Superoxide Dismutase

The role of superoxide dismutase in the body

Superoxide dismutase (SOD) is one of the body's primary internal antioxidant defences, and plays a critical role in reducing the oxidative stress implicated in atherosclerosis and other life-threatening diseases.

SOD catalyses the dismutation of superoxide into oxygen O_2 and hydrogen peroxide H_2O_2 .



Superoxide O_2^- is one of the most dangerous free radicals present in the human body and by destroying it, the SOD is performing a vital function.

SOD is involved in lipid peroxidation (the process in which free radicals steal an electron from lipids in the cell membrane, resulting in cellular damage). Most aerobic cells contain the SOD enzyme.

SOD helps regulate the production of antioxidant enzymes, including catalase (CAT) and glutathione peroxidase (GPx), which are required to neutralize reactive oxygen species (ROS).

Applications

Clinical

- SOD deficiency is associated with diseases such as the familial dominant form of amyotrophic lateral sclerosis (ALS) and Alzheimer's disease
- SOD can be used to treat various ailments including arthritis, burns and inflammatory diseases. Its measurement can help to evaluate the efficiency of treatment targeting SOD levels.

Research

- To determine the therapeutic efficacy and antioxidant potential of newly developed drugs.
- To research the association of SOD with a number of disease states.

Sports

- The SOD test allows for research into the effects of sport on free radical damage and the protective effects of SOD.

Normal Reference Range: 130-222 U/mL

Promo RPP: RM80

Validity till 30/6/2026

TAT: 3-5 working days

Sample Requirements:

1. Sample type: 3ml fresh hemolyfree plain serum or EDTA anticoagulant plasma. After specimen collection, avoid direct sunlight and timely seal.
2. Sample preservation: 2°C - 8°C, sample to be collected by dispatch rider on the same day.

Professional Information Sheet for Doctors

Clinical Significance

Superoxide Dismutase (**SOD**) is an essential antioxidant enzyme that protects against cellular oxidative damage caused by reactive oxygen species (ROS). Low SOD activity reflects diminished antioxidant defense and can be associated with **aging, chronic inflammation, diabetes, neurodegeneration, and metabolic disorders**.

When to Recommend the Test

- Patients with **chronic inflammation, diabetes, or metabolic disorders**
- **Neurodegenerative diseases** (e.g., Alzheimer's, Parkinson's)
- **Autoimmune or oxidative-stress-related conditions**
- **Lifestyle risks** – smoking, athletes, toxin exposure
- To complement **antioxidant profiling** (CAT, GPx, TAC)

Interpretation Summary

SOD (U/mL)	Interpretation	Possible Causes / Notes	Next Steps
> 222 (High)	Enhanced enzymatic activity	Exercise, acute inflammation, stress	Assess hs-CRP, monitor recovery
130 – 222 (Normal)	Balanced oxidative state	Healthy or stable chronic condition	Re-test every 6–12 months
< 130 (Low)	Depleted antioxidant enzyme reserve	Aging, diabetes, hepatic/neuro disorders	Add FNAVA (Vit. E/C, CoQ10, ALA)

Clinical Algorithm

1. Identify patient with oxidative stress risk factors
2. Order **SOD Test**
3. Interpret based on thresholds
4. Correlate with clinical and lifestyle findings
5. Initiate antioxidant/lifestyle intervention
6. Re-test after **6–8 weeks**

Management Guidance

For Low SOD

- **Diet:** Add antioxidant-rich foods (greens, berries, citrus, nuts, legumes)
- **Micronutrients:** Ensure zinc, copper, manganese
- **Supplements:** Consider SOD-mimetic formulas, CoQ10, Vitamin C/E, ALA
- **Clinical:** Address underlying inflammation/metabolic cause

For High SOD

- Often compensatory; assess acute inflammation or physical stress
- Re-test post-recovery phase

Report Remark

Superoxide Dismutase (SOD) activity was measured as an index of the body's antioxidant defense. Low values indicate reduced enzymatic protection against superoxide radicals and potential oxidative stress burden. Clinical correlation with lifestyle, nutritional, and metabolic status is recommended.